“PAPER_{0:D3}” -f [int]# PAPER #82 □ Black Hole Evaporation Timescales: UQFF Corrections

Title: UQFF-Corrected Black Hole Evaporation Timescales: Stellar Mass Through Primordial Black Holes

Author: Daniel T. Murphy
Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)
Date: March 7, 2026
Source Data: validate_hawking_temperature.py (Tests 2, 6), CondensedPhysics.py simulate_evaporation()
Index Slot: §1.11 Black Hole Physics & Hawking Radiation,
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Index Slot: §1.11 Black Hole Physics & Hawking Radiation, PAPER_082

Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result t_evap = 5120pG^{2}M^3/(?c4) is modified by the T_UQFF/T_H = 0.99 ratio. This changes evaporation rates by a factor of (T_UQFF/T_H)4 = 0.96, extending evaporation timescales by ~4% for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10$^{1°}$ kg over 100 timesteps.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Evaporation Timescale Formula

Standard GR

$$t_{\text{evap}}^{\text{GR}} = \frac{5120\pi G^2 M_0^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\text{evap}}^{\text{UQFF}} = \frac{t_{\text{evap}}^{\text{GR}}}{(T_{\text{UQFF}}/T_H)^4} = \frac{t_{\text{evap}}^{\text{GR}}}{0.99^4} = t_{\text{evap}}^{\text{GR}} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR □ black holes are slightly more stable in the UQFF vacuum.

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2. Evaporation Timescales: Full Table

System	M0	t_evap_GR	t_evap_UQFF	Survives Universe
Sgr A*	4×106 M?	8.7×108^3 s	9.1×108^3 s	? Yes
M87*	6.5×10? M?	3.8×10?5 s	4.0×10?5 s	? Yes
Stellar BH	10 M?	2.1×1074 s	2.2×1074 s	? Yes
Primordial BH	5.7×10^{11} kg	$4.35\times10^{1}7$ s = t_U	$4.52\times10^{1}7$ s	Borderline
Primordial BH	$1\times10^{1°}$ kg	2.3×10^{12} s (73 kyr)	2.4×10^{12} s	? Evaporated

The validate_hawking_temperature.py Test 2 confirms: - Stellar BH (10 M?): `survives_universe = True` ? - Test 6 simulation: M_initial = $10^{1°}$ kg, 100 steps, mass_lost_fraction computed ?

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3. Mass Evolution Simulation

From `simulate_evaporation(M_initial = 10¹° kg, dt = 10¹° s, n_steps = 100)`:

$$\frac{dM}{dt} = -\frac{k_{\mathrm{UQFF}}}{M^2}, \quad k_{\mathrm{UQFF}} = \frac{\hbar c^4 (T_{\mathrm{UQFF}}/T_H)^4}{15360\pi G^2}$$

With T_UQFF/T_H = 0.99: k_UQFF = 0.96 × k_GR

At t = 100 × $10^{1°}$ s = 10^{12} s: - M_final M_initial × (1 - t/t_evap)^{1/3} = $10^{1°}$ × (1 - $10^{12}/2.4\times10^{12}$)^{1/3} - M_final $10^{1°}$ × 0.583^{1/3} 8.35×10? kg - Mass lost fraction **16.5%** over first 10^{12} s

Arrays `times[]`, `masses[]`, `temperatures_H[]` all have matching lengths (validate Test 6). ?

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4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation (M ? M_Planck):

$$\frac{dM_{\mathrm{UQFF}}}{dt} = -\frac{k_{\mathrm{UQFF}}}{M^2} + \frac{g_{\mathrm{Buoyant}} \times V_{\mathrm{BH}}}{c^2}$$

The buoyancy term: g_Buoyant × V_BH / c^2 = ?_vac × 1055 × (4/3)p r_S^3 / c^2 ~ 10?8° kg/s ? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_GR	0.96 × k_GR	-4%
Timescale t_evap	t_GR	1.041 × t_GR	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	5.7×10^{11} kg	5.5×10^{11} kg	-3.5%
Test 2	`survives = True`	Confirmed	? PASS

Source: validate_hawking_temperature.py Tests 2+6, simulate_evaporation() | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER_{0:D3}" -f $n

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Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result t_evap = 5120pG²M³/(?c4) is modified by the T_UQFF/T_H = 0.99 ratio. This changes evaporation rates by a factor of (T_UQFF/T_H)4 = 0.96, extending evaporation timescales by ~4% for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10¹° kg over 100 timesteps.

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UQFF evaporation timescale is 4.1% longer than GR black holes are slightly more stable in the UQFF vacuum.

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Source: validate_hawking_temperature.py Tests 2+6, simulate_evaporation() | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value Black Hole Evaporation Timescales: UQFF Corrections

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Index Slot: §1.11 Black Hole Physics & Hawking Radiation, PAPER_082

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Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result t_evap = 5120pG^2M^3/(?c4) is modified by the T_UQFF/T_H = 0.99 ratio. This changes evaporation rates by a factor of (T_UQFF/T_H)4 = 0.96, extending evaporation timescales by ~4% for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at $10^{1\circ}$ kg over 100 timesteps.

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From `simulate_evaporation(M_initial = 10¹° kg, dt = 10¹° s, n_steps = 100)`:

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2. Evaporation Timescales: Full Table

System	M0	t_evap_GR	t_evap_UQFF	Survives Universe
Sgr A*	4×106 M?	8.7×108³ s	9.1×108³ s	? Yes
M87*	6.5×10? M?	3.8×10?5 s	4.0×10?5 s	? Yes
Stellar BH	10 M?	2.1×1074 s	2.2×1074 s	? Yes
Primordial BH	5.7×10¹¹ kg	4.35×10¹7 s = t_U	4.52×10¹7 s	Borderline
Primordial BH	1×10¹° kg	2.3×10¹² s (73 kyr)	2.4×10¹² s	? Evaporated

The validate_hawking_temperature.py Test 2 confirms: - Stellar BH (10 M?): `survives_universe = True` ? - Test 6 simulation: M_initial = 10¹° kg, 100 steps, mass_lost_fraction computed ?

---

3. Mass Evolution Simulation

From `simulate_evaporation(M_initial = 10¹° kg, dt = 10¹° s, n_steps = 100)`:

$$\frac{dM}{dt} = -\frac{k_{\mathrm{UQFF}}}{M^2}, \quad k_{\mathrm{UQFF}} = \frac{\hbar c^4 (T_{\mathrm{UQFF}}/T_H)^4}{15360 \pi G^2}$$

With T_UQFF/T_H = 0.99: k_UQFF = 0.96 × k_GR

At t = 100 × 10¹° s = 10¹² s: - M_final □ M_initial × (1 - t/t_evap)^{1/3} = 10¹° × (1 - 10¹²/2.4×10¹²)^{1/3} - M_final □ 10¹° × 0.583^{1/3} □ 8.35×10? kg - Mass lost fraction □ **16.5%** over first 10¹² s

Arrays `times[]`, `masses[]`, `temperatures_H[]` all have matching lengths (validate Test 6). ?

---

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation (M ? M_Planck):

$$\frac{dM_{\mathrm{UQFF}}}{dt} = -\frac{k_{\mathrm{UQFF}}}{M^2} + \frac{g_{\mathrm{Buoyant}} \times V_{\mathrm{BH}}}{c^2}$$

The buoyancy term: g_Buoyant × V_BH / c² = ?_vac × 1055 × (4/3)p r_S³ / c² ~ 10?8° kg/s ? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_GR	0.96 × k_GR	-4%
Timescale t_evap	t_GR	1.041 × t_GR	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	5.7×10^{11} kg	5.5×10^{11} kg	-3.5%
Test 2	`survives = True`	Confirmed	? PASS

Source: validate_hawking_temperature.py Tests 2+6, simulate_evaporation() | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value

---

Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result t_evap = 5120pG²M³/(?c4) is modified by the T_UQFF/T_H = 0.99 ratio. This changes evaporation rates by a factor of (T_UQFF/T_H)4 = 0.96, extending evaporation timescales by ~4% for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10¹° kg over 100 timesteps.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?¹, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. Evaporation Timescale Formula

Standard GR

$$t_{\text{evap}}^{\text{GR}} = \frac{5120\pi G^2 M_0^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\text{evap}}^{\text{UQFF}} = \frac{t_{\text{evap}}^{\text{GR}}}{(T_{\text{UQFF}}/T_H)^4} = \frac{t_{\text{evap}}^{\text{GR}}}{0.99^4} = t_{\text{evap}}^{\text{GR}} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR black holes are slightly more stable in the UQFF vacuum.

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Source: validate_hawking_temperature.py Tests 2+6, simulate_evaporation() | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER_{0:D3}" -f $n

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$$\frac{dM}{dt} = -\frac{k_{\text{UQFF}}}{M^2}, \quad k_{\text{UQFF}} = \frac{\hbar c^4 (T_{\text{UQFF}}/T_H)^4}{15360\pi G^2}$$

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Source: validate_hawking_temperature.py Tests 2+6, simulate_evaporation() | ? = 0.0005/day | [SSq] = 0.57

UQFF computed: Eddington luminosity UQFF correction = 1 - [SSq]×exp(-?×?t) = 1 - 5.7e-1 × exp(-2.9e-4) = 4.3e-1; F_U at event horizon = 2.0e+18 m/s².

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
Îº	5.0 Ã 10â»â´ dayâ»Â¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
Î²_i	0.60â0.61	Buoyancy coupling coefficient
kâ	1.5	Ug1 DPM-dipole coupling
kâ	1.2	Ug2 outer-bubble charge coupling
kâ	1.8	Ug3 string-rotation coupling
kâ	2.0	Ug4 vacuum-concentration coupling
Î·	10â»Â²Â²	Inertia tensor scale
E_react(0)	10â´â¶ J	Reference reactive energy

A.2 F_U Master Equation (Complete â 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} igl[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}} igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
âÎ£Î»áµ¢Â·Uáµ¢Â·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
Î»â=10â»Â¹â°, Î»â=10â»Â¹Â², Î»â=10â»Â¹Â¹, Î»â=10â»Â¹Â³ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13}\,\Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q \cos(\Delta\omega\, t)igr)$$

Symbol	Value	Description
Ï_c	10Â¹âµ kg/mÂ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
ÎÏ	2Ï/(434Â·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, â□¦)	Multi-scale field interactions
Buoyant	Î²_i Ã□ Ubi	Expanding nebulae, stellar winds
Superconductive	Um Ã□ (1+10Â¹Â³Â·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.